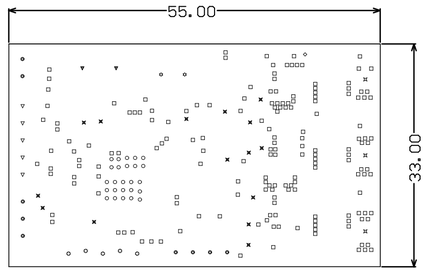
REV	ECO	Comments	Date
1	–	Initial Design	8/15/2024
2	–	Revised Layout	01/30/2026



Symbol	Hit Count	Finished Hole Size	Plated	Via/Pad	Hole Tolerance
□	174	0.250mm (9.84mil)	PTH	(Mixed)	+0/-0.100mm (+0/-3.94mil)
○	25	0.300mm (11.81mil)	PTH	Via	+0/-0.100mm (+0/-3.94mil)
✖	15	0.400mm (15.75mil)	PTH	Pad	+/-0.076mm(+/-3mil)
◇	1	0.600mm (23.62mil)	PTH	Pad	+/-0.076mm(+/-3mil)
✡	2	0.800mm (31.50mil)	PTH	Pad	+/-0.076mm(+/-3mil)
◎	9	0.900mm (35.43mil)	PTH	Pad	+/-0.076mm(+/-3mil)
▽	5	1.020mm (40.16mil)	PTH	Pad	+/-0.076mm(+/-3mil)
✪	5	1.067mm (42.00mil)	PTH	Pad	+/-0.076mm(+/-3mil)
⌘	3	1.300mm (51.18mil)	PTH	Pad	+/-0.076mm(+/-3mil)
▼	2	2.540mm (100.00mil)	PTH	Pad	+/-0.076mm(+/-3mil)
	241 Total				

THIS PCB TO BE MANUFACTURED TO MEET ALL ACCEPTANCE LEVELS OF A CLASS 2 PCB PER ANSI/IPC-A-600G.

MATERIAL: FR-4 (Tg170) or Equivalent

- [X] MULTILAYER [4] LAYERS [ ] CONTROLLED IMPEDANCE
- Cu WEIGHT EXTERNAL LAYERS [70um] FINISHED
- Cu WEIGHT INTERNAL LAYERS [35um] FINISHED
- FINISHED OVERALL THICKNESS [1.6] mm ± [10] %
- COPPER THIEVING ALLOWED [ ] YES [X] NO

FINISH:

- [ ] LEAD-FREE HOT AIR LEVELING
- [X] IMMERSION GOLD
- [ ] IMMERSION TIN
- [ ] SMOBC WITH SELECTIVE GOLD PLATING ON LANDS INDICATED. 1um GOLD OVER 5-10 um NICKEL

SOLDERMASK: DYNACHEM EPIC 200 LPI OR EQUIVALENT

SOLDERMASK COLOR: RED HIGH GLOSS

SILKSCREEN COLOR: WHITE

DRILLING: [ ] Blind / Burried Vias

Via Layer Pairs: Top-Bottom

ALL HOLES TO BE LOCATED BY THE COORDINATES FROM THE NC DRILL DATA PROVIDED.

USE ARTWORK SET NO. 04-12307 PCB REV 2

ALL UL LOGO, MANUFACTURER'S ID, AND DATE CODES SHALL BE PLACED ON THE AVAILABLE SPACE ON THE BOTTOM SIDE SILKSCREEN LAYER UNLESS OTHERWISE INDICATED

ANY ALTERNATIVES TO THE ABOVE SPECIFICATIONS MUST FIRST BE APPROVED.

REFERENCE STACK-UP (ONLY):

Layer	Name	Material	Thickness
	Top Overlay		
	Top Solder	Solder Resist	0.010mm
1	LAYER 1 (Top)		0.035mm
	Dielectric 1	FR-4	0.453mm
2	LAYER 2		0.035mm
	Dielectric 2	FR-4	0.457mm
3	LAYER 3		0.035mm
	Dielectric 3	FR-4	0.453mm
4	LAYER 4 (Bottom)		0.035mm
	Bottom Solder	Solder Resist	0.010mm
	Bottom Overlay		
Total board thickness:			1.524mm

MICROCHIP
Microchip Technology Inc.
2355 W. Chandler Blvd.
Chandler, AZ 85224

TITLE: dsPIC33CDVC Drone ESC Reference Design

PART NUMBER: 12307

PCB DESIGNER:	GERBER FILE:		
S. Balong Jr.	'Fab Drawing (Bottom,Top Layer)'		
ENGINEER:	BOARD NUMBER:	DOCUMENT NUMBER:	DATE:
Loida Canada	04-12307	04-12307-D	1/30/2026
PCB FILE NAME:	LAYER NAME:	REV:	
04-12307_Drone_ESC.PcbDoc	FAB (M4) Drill Drawing	2	